CHAPTER 01
INTRODUCTION

1.1 BACKGROUND OF THE PROBLEM

Livestock conformation assessment is a critical part of dairy cattle breeding, productivity assessment, and health evaluation. Body conformation measurements such as body length, chest measurement, and rump type are commonly used in Animal Type Classification (ATC) systems to assess functional efficiency and longevity of cattle. Traditional body conformation assessment techniques are carried out manually by trained personnel. However, manual assessment is susceptible to human error, time-consuming, and subjective, particularly when carried out in a field environment.

Recent advances in computer vision and deep learning have made it possible to automate the analysis of animal body morphology from images. The pose estimation models possess the capability to identify points of interest on animals, which makes it possible to calculate geometric parameters without the need for direct access. This opens up opportunities for the development of objective, accurate, and scalable ATC systems for image analysis only.

1.2 MOTIVATION FOR THE PROJECT

Manual cattle conformation evaluation is a process that requires experienced personnel and identical evaluation conditions, which are not always possible in a rural or field setting. Inconsistent evaluation will impact breeding decisions. An automated image-based ATC system can remove human bias, save time, and offer fast evaluation in a farm setting. The purpose of this project is to develop a functional computer vision system that can estimate cattle structural characteristics and ATC scores from a single color image.

1.3 PROBLEM DEFINITION / STATEMENT

Animal Type Identification via hand, performed by trained animal typers is subjective, prone to error from fatigue and visual bias errors. Measurement can also be very inaccurate. Inconsistencies like these impact data quality with potential influence on selection of elite dams under national breeding programs like Rashtriya Gokul Mission. For this to be possible, video-based evidence can provide a solution by removing subjectivity while allowing for large-scale data collection under heterogeneous field conditions, but an automated, objective and scalable pipeline for reliable assessment of

animal body structure and assign standardized ATC scores with minimal human intervention.

1.4 OBJECTIVES OF THE PROJECT

The primary objectives of this project are:

- To automate the Animal Type Classification process for cattle and buffaloes using AI and image processing.
- To reduce subjectivity, bias, and human error present in manual ATC methods.
- To accurately detect anatomical landmarks and extract key body structure parameters.
- To generate standardized and reproducible ATC scores.
- To support scientific decision-making in breeding programs under PT and PS schemes.

1.5 SCOPE OF THE SYSTEM

The proposed system concentrates on the estimation of structural conformation traits based on images of cattle taken from the side. The system is intended for the assessment of dairy cattle morphology and ATC scoring in real-world conditions on a farm. The system does not estimate weight based on measurements or classify types of cattle. The system estimates scores of structural traits based on the geometry of anatomical landmarks.

1.6 METHODOLOGY OVERVIEW

The proposed methodology consists of four major steps:

- Image Acquisition: The cattle images are obtained from a labeled pose dataset that holds anatomical keypoints.
- Landmark Detection: A YOLOv8-Pose deep learning algorithm is employed to identify the anatomical keypoints on the cattle images.
- Morphometric Ratio Computation: The distance between key keypoints is calculated to determine structural ratios based on withers height.
- ATC Scoring: The structural ratios are normalized to determine standardized ATC scores that represent body length, chest capacity, and rump length.
- The system returns trait scores and an ATC index for each cattle image.

CHAPTER 02
LITERATURE REVIEW

2.1 OVERVIEW OF PREVIOUS RESEARCH

Automated cattle analysis has progressed from conventional image processing techniques to deep learning-based techniques. Conventional techniques relied on image segmentation and edge detection to determine the height and length of the cattle from images, but these techniques were very light-dependent and prone to complex backgrounds. The development of convolutional neural networks improved the cattle detection system in a farm setting using techniques such as Faster R-CNN and YOLO. Nevertheless, these detection techniques were mainly used for cattle detection and not for determining anatomical dimensions necessary for structural analysis.

Current research has progressed towards pose estimation, where deep learning algorithms identify keypoint locations in cattle images for purposes of posture analysis and lameness detection.Keypoint-based methods allow for non-contact measurement of geometric dimensions such as withers height and body length, making morphometric analysis feasible. Although considerable progress has been made, most recent studies are focused on predicting the value of body parameters instead of using keypoints to predict standardized conformation scores. There is, therefore, a clear need for the development of automated Animal Type Classification (ATC) scoring pipelines from keypoints.

2.2 COMPARISON OF EXISTING METHODS AND MODELS

Paper / Study	**Technique Used**	**Dataset Type**	**Measurements Extracted**	**ATC / Conformation Scoring**	**Limitations**
Mirzaei et al., “Vision-Based Measurement of Cattle Body Dimensions”	Classical image processing	Controlled cattle images	Body length, height	No	Sensitive to lighting and background
Ren et al., “Faster R-CNN: Towards Real-Time Object Detection”	CNN object detection	General object datasets	Bounding box only	No	No anatomical measurements

Bochkovskiy et al., "YOLOv4: Optimal Speed and Accuracy of Object Detection"	CNN object detection	Farm animal images	Bounding box only	No	No structural traits
Viazovetskyi et al., "Depth-Based Estimation of Livestock Body Size"	RGB-D sensing	Barn depth images	Absolute dimensions	No	Requires depth camera
Mathis et al., "DeepLabCut: Markerless Pose Estimation of Animals"	Keypoint detection	Animal pose datasets	Landmark coordinates	No	Not cattle-specific ATC
Wu et al., "Multi-Animal Pose Estimation in Cattle"	Pose estimation	Farm cattle images	Joint positions	No	Behavior-focused
Huang et al., "Image-Based Estimation of Cattle Body Size"	CNN regression	Labeled cattle dataset	Body length, girth	No	No conformation scoring
Proposed YOLOv8-Pose ATC System (This Work)	Pose + morphometric ratios	Cattle pose dataset	OBL/WH, HG/WH, HL/WH	Yes	Depends on visible landmarks

2.3 LIMITATIONS AND RESEARCH GAPS IDENTIFIED

Despite the progress in livestock pose estimation, there are still some limitations in the existing research. Most of the research is focused on animal posture or behavior detection instead of structural conformation assessment. Measurement-based research is

focused on absolute body measurements without normalization to standardized scoring systems used in breeding assessment.

Most of the existing research requires controlled environments, depth cameras, or calibration markers, which are not applicable in real-world farm environments. Moreover, there are very few research studies that cover the entire pipeline from landmark detection to morphometric ratio calculation and conformation scoring. As a result, the research on automatic ATC-style scoring from RGB images is not adequately explored yet.

Another limitation of the existing research is the absence of geometric stability and consistency analysis of landmark-based measurements. Most of the pose estimation research studies are focused on detection accuracy without considering the reliability of morphometric ratios for structural analysis.

2.4 HOW YOUR WORK DIFFERS UPON PREVIOUS STUDIES

The proposed research work bridges the gaps by developing an integrated image-based cattle conformation assessment system through pose estimation. The YOLOv8-Pose model is used to detect the key anatomical points of the cattle directly from the RGB images captured in a natural farm environment without using any markers or depth cameras.

Morphometric ratios for structural characteristics like body length, chest capacity, and rump length are calculated with respect to the withers height, making it scale-independent for conformation evaluation. The morphometric ratios are then converted to standardized ATC scores, providing a quantitative structural evaluation aligned with the livestock assessment tradition.

In contrast to existing works that concentrate on measurement prediction or pose estimation alone, the proposed system combines landmark detection, geometric measurement, and ATC scoring into a single framework. The system also assesses the landmark robustness and score variation within a dataset, establishing the validity of the extracted morphometric information.

The proposed system allows for the automatic estimation of the ATC score from a single image, providing an objective and efficient means for manual conformation assessment.

CHAPTER 03
METHODOLOGY

3.1 PROBLEM OVERVIWE

Structural conformation evaluation in cattle is a critical task for assessing productivity, functional efficiency, and breeding value. The conventional scoring of Animal Type Classification (ATC) is carried out by manual measurement and assessment of anatomical features such as body length, chest capacity, and rump form.

However, with the development of computer vision technology, it has become possible to automatically identify the anatomical points of cattle in images using pose estimation models. The anatomical points can be used to identify the coordinates of the critical points on the body, which can be measured geometrically without the need for physical contact. However, the current cattle pose estimation system is only capable of posture analysis or cattle observation and is not equipped with the capability to link these points to standardized conformation scores.

The problem addressed in this research is the design of an automated image processing system that can map cattle anatomical landmarks localized from images to morphometric ratios and standardized ATC conformation scores. The system should be capable of processing RGB images taken in natural farm settings and be able to provide accurate structural trait evaluation without the need for specialized equipment or manual measurement.

3.2 DATASET DESCRIPTION

The dataset is comprised of images of dairy cattle captured in real-world environments with different lighting conditions, background, pose, orientation, and distance from the camera. Each image contains a single cattle with 26 keypoints marked on the body, which includes the head, torso, limbs, withers, hip, and tail base. The keypoints are used to determine the two-dimensional coordinates of the body landmarks, which are critical in pose estimation modeling.The dataset is divided into training, validation, and testing sets for model development and testing.

In contrast to conventional livestock image datasets that record manual body measurements, this pose-annotated dataset enables the direct computation of geometric body dimensions from detected keypoints. The availability of detailed anatomical landmarks makes it suitable for automated morphometric analysis and structural

conformation evaluation from images, supporting the development of computer vision–based cattle assessment systems without physical contact.

The dataset was divided as follows:

- Training set: 70%
- Validation set: 20%
- Test set: 10%

3.3. DATA PREPROCESSING

The images and annotations are preprocessed before training and testing to ensure compatibility with the YOLOv8-Pose model and to provide consistent morphometric characteristics.

3.3.1 Image Preprocessing

The images are resized to a fixed resolution to facilitate efficient training and testing of the model. The RGB color channels are maintained to preserve visual cues essential for precise keypoint localization.

3.3.2 Annotation Formatting

The keypoint annotations are formatted according to the YOLOv8-Pose convention, including normalized coordinates and visibility masks for each keypoint. This standardization enables the model to effectively learn spatial relationships between anatomical landmarks.

3.3.3 Data Augmentation

Horizontal flipping, scaling, and brightness adjustments are applied during training to simulate variations in orientation, distance, and illumination conditions encountered in farm environments.

3.3.4 Landmark Selection for Morphometry

From the 26 annotated keypoints, only those relevant to structural conformation are selected to compute body length, chest capacity, and rump length.

3.3.5 Scale Normalization

Geometric distances are normalized by withers height to remove the influence of camera distance. This ensures that morphometric characteristics reflect true body proportions rather than pixel-based measurements.

3.4. METHODOLOGY OVERVIEW

The proposed methodology uses a landmark-based morphometric analysis method that consists of four steps: landmark detection, geometric measurement, ratio normalization, and ATC scoring.

3.4.1 Landmark Detection

The anatomical landmarks in the cattle images are identified by a deep learning pose estimation technique, YOLOv8-Pose. The pose estimation technique estimates the coordinates of a set of keypoints that define the anatomical points such as withers, shoulder, hip, rump, and torso. The keypoints define the geometric form of the cattle body, which is required for morphometric analysis.

Given an input image I, the pose model produces a set of keypoints:

$$\boxed{K = \{(x_i, y_i)\}_{i=1}^{N}}$$

where N is the number of anatomical landmarks detected on the cattle body.

3.4.2 Geometric Measurement Extraction

Morphometric measurements are derived from Euclidean distances between selected anatomical landmarks. The following structural measurements are computed:

Oblique Body Length (OBL):

Distance between shoulder and hip landmarks.

$$\boxed{OBL = \| K_{shoulder} - K_{hip} \|}$$

Withers Height (WH):

Vertical distance between withers and ground landmarks.

$$\boxed{WH = \| K_{withers} - K_{ground} \|}$$

Hip Length (HL):

Distance between hip and tail-base landmarks.

$$\boxed{HL = \| K_{hip} - K_{tail} \|}$$

Heart Girth (HG):

Horizontal torso width between left and right body landmarks.

$$\boxed{HG = \| K_{left} - K_{right} \|}$$

3.4.3 Morphometric Ratio Normalization

Absolute measurements depend on animal size and image scale. To obtain scale-independent structural traits, measurements are normalized relative to withers height. The following ratios are computed:

$$\boxed{R_{OBL} = \frac{OBL}{WH}, R_{HG} = \frac{HG}{WH}, R_{HL} = \frac{HL}{WH}}$$

These ratios represent body proportions rather than absolute size, enabling consistent comparison across animals and images.

3.4.4 ATC Score Computation

Morphometric ratios are mapped to standardized ATC conformation scores using biologically defined ranges for each structural trait. Each ratio is normalized to a score between 0 and 1:

$$S = \frac{R - R_{min}}{R_{max} - R_{min}}$$

where R is the measured ratio and R_{min}, R_{max} represent expected biological limits.

The following ATC trait scores are computed:

- Body Length Score
- Chest Capacity Score
- Rump Length Score

The overall ATC index is calculated as the mean of the three trait scores:

$$ATC = \frac{S_{BodyLength} + S_{ChestCapacity} + S_{RumpLength}}{3}$$

This index represents overall structural conformation of the cattle.

3.5 TECHNIQUES AND MODELS USED

The proposed system integrates pose estimation with deep learning methods and geometric morphometric analysis.

1. **YOLOv8-Pose Model:**
   A real-time object detection and keypoint estimation model for the detection of cattle anatomical points from RGB images. The model has the ability to accurately locate keypoints with low inference latency.
2. **Morphometric Ratio Analysis**:
   The distances between anatomical points are measured using Euclidean geometry. Normalization with respect to withers height enables scale-independent structural analysis.
3. **ATC Scoring Model:**
   A rule-based normalization strategy normalizes morphometric ratios to standardized conformation scores based on biologically valid ranges ascertained from literature on livestock morphology.
4. **Dynamic Input Interface:**
   A user-interaction module enables the user to upload cattle images and automatically calculate ATC scores, thus demonstrating the end-to-end automated conformation analysis process.

3.6. EVALUATION METRICS

The performance of the proposed system is evaluated based on the accuracy and consistency of morphometric measurements and corresponding ATC scores obtained from identified anatomical landmarks.

3.6.1 Morphometric Measurement Accuracy :

Structural measurements such as body length, chest capacity, and rump length are derived from predicted anatomical landmarks and validated against reference measurements. Accuracy is assessed using Mean Absolute Error (MAE) and percentage error to quantify geometric precision.

3.6.2 Ratio Consistency after Normalization :

All structural measurements are normalized by withers height to obtain scale-independent body proportions. The stability of these normalized ratios across images with varying cattle distance and size is evaluated using variance analysis. Low variance values indicate effective scale normalization.

3.6.3 ATC Score Reliability :

Animal Type Classification (ATC) scores derived from normalized morphometric ratios are evaluated for consistency across the dataset. Stable score distributions and logical trait relationships demonstrate that the automated scoring system preserves structural conformation characteristics.

3.6.4 End-to-End Pipeline Robustness :

The complete pipeline is evaluated across different cattle orientations and farm backgrounds. Successful generation of valid measurements and scores across samples indicates the practical robustness of the automated conformation assessment system.

3.7 FLOWCHART / SYSTEM ARCHITECTURE

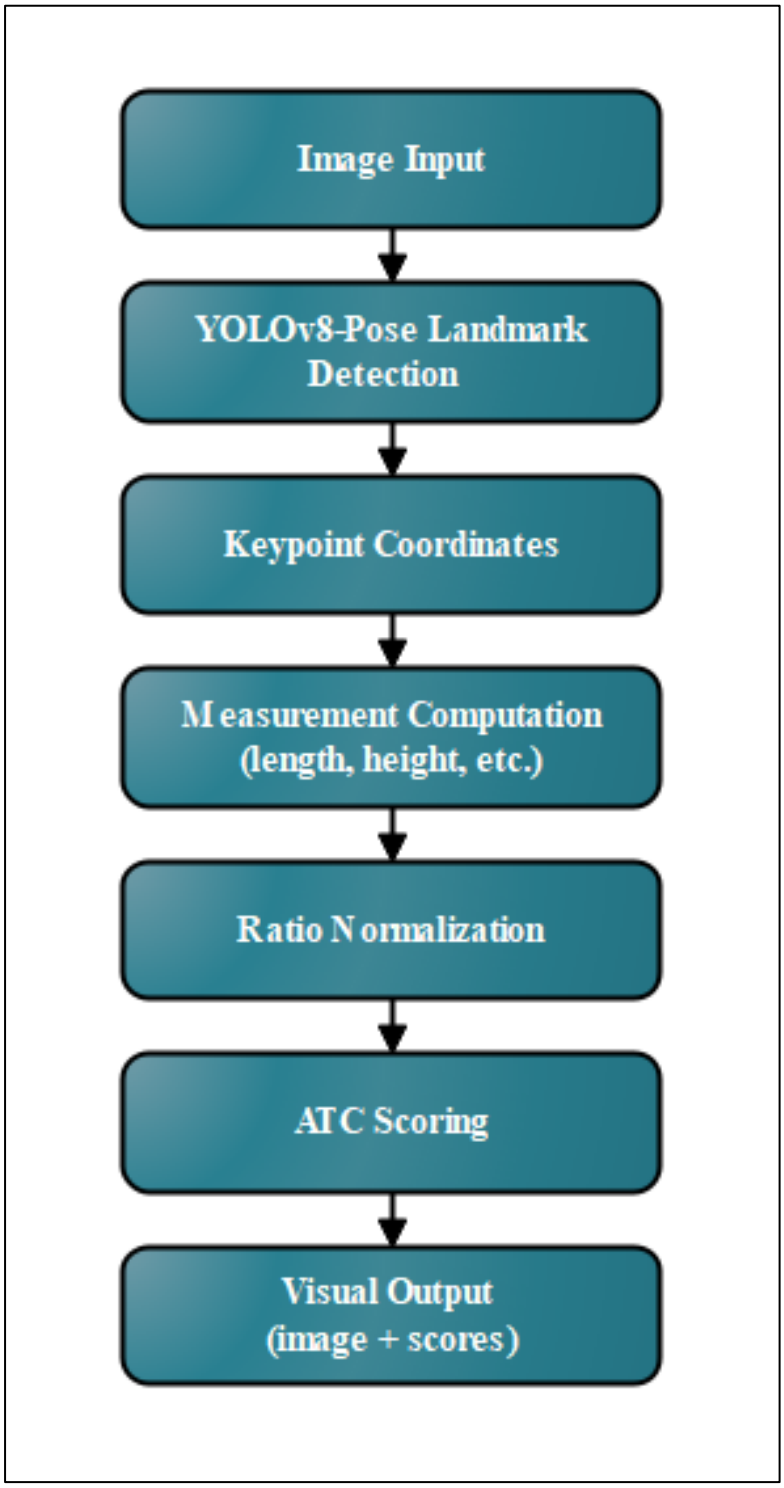

Fig no. 3.1 FLOWCHART

Figure No.3.1:Self-desugned flowchart illustrating the Cattle measurement system.
(Source : Self Generated)

CHAPTER 04

IMPLEMENTATION AND RESULTS

4.1 EXPERIMENTAL SETUP AND IMPLEMENTATION DETAILS

The cattle ATC scoring model was created using Python programming and deep learning techniques. The system was evaluated in a cloud-based environment using Google Colab with GPU support to ensure efficient pose estimation inference. The YOLOv8-Pose model was utilized to detect anatomical keypoints in cattle images, and morphometric ratios were calculated from these detected landmarks to generate ATC scores.

The pose estimation model was tested on the designated test subset of the cattle pose dataset. For each image, anatomical keypoints were automatically detected and geometric measurements were computed. Structural ratios representing body length, chest capacity, and rump length were normalized to obtain ATC scores, and a composite ATC index was derived. The evaluation assessed detection accuracy, geometric consistency, score variability, and computational efficiency of the overall system.

4.2 DATASET USED FOR TRAINING

The evaluation experiments were conducted on the test set of the cattle pose dataset, which contains cattle images annotated with cattle images taken in natural farm settings. The images vary in cattle orientation, size, and background complexity. Each image is annotated with landmarks of the cattle, which makes it possible to extract geometric measurements.

Only images in which the cattle landmarks were successfully detected by the cattle pose model were used for morphometric analysis. The dataset has enough variation in cattle morphology to test the discriminative power of the proposed ATC scoring system.

4.3 PERFORMANCE EVALUATION METRICS

The performance of the proposed system was evaluated using relevant measures that are more closely related to the needs of landmark-based morphometric analysis than classification accuracy. The following measures were used:

1. **Detection Rate:**

The rate at which images were successfully detected with cattle landmarks by the pose model.

$$Detection\ Rate = \frac{\text{Detected Images}}{\text{Total Images}}$$

2. **Morphometric Ratio Stability:**

The standard deviation of structural ratios (OBL/WH, HG/WH, HL/WH) in the dataset. A low standard deviation indicates that the landmarks are properly localized and the geometric ratios are properly measured.

3. **Trait Score Variability:**

The standard deviation of the normalized trait scores (Body Length, Chest Capacity, Rump Length). A non-zero standard deviation indicates that the system has the capability to differentiate structural conformation differences among cattle.

4. **ATC Index Variability:**

The standard deviation of the overall ATC index values among the cattle images. This measures the capability of the scoring system to differentiate.

5. **Inference Time:**

The average time taken for landmark detection and ATC computation per image, which indicates the computational efficiency of the system and its capability to operate in real-time.

4.4 RESULTS AND ANALYSIS

The YOLOv8-Pose model was able to correctly identify anatomical points in almost all test images, with a detection rate close to 1.0, thus exhibiting high robustness in a natural farm environment. The calculated morphometric ratios (OBL/WH, HG/WH, HL/WH) had a low standard deviation among samples, thus ensuring consistent anatomical point identification. This thus validates that the pose estimation module is able to identify accurate spatial information necessary for structural conformation analysis.

The ATC trait scores on the normalized ATC had biologically significant variation among cattle, with body length, chest capacity, rump length, and total ATC index varying based on structural conformation. This thus validates the system's capability to identify and distinguish among cattle based on morphology. The average end-to-end processing time of 10-20 ms per image on GPU is thus sufficient for real-time processing.

4.5 SCREENSHOTS

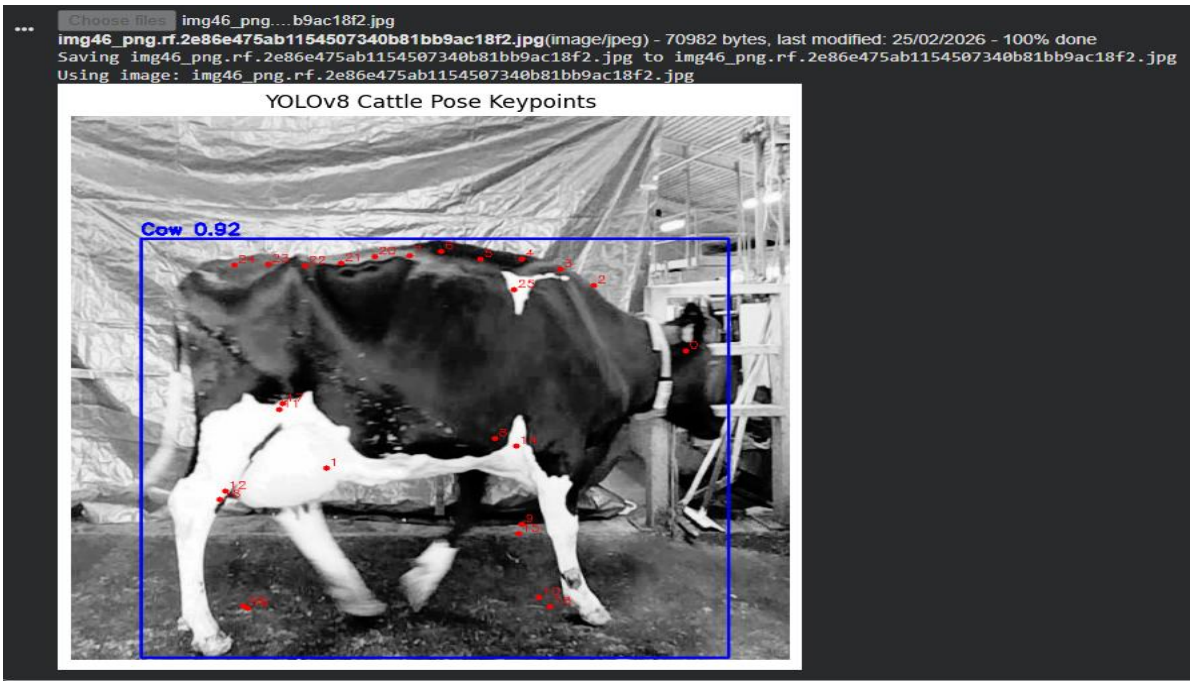

Fig no. 4.6 Landmark Detection
(Source : Our Code Output Screenshot)

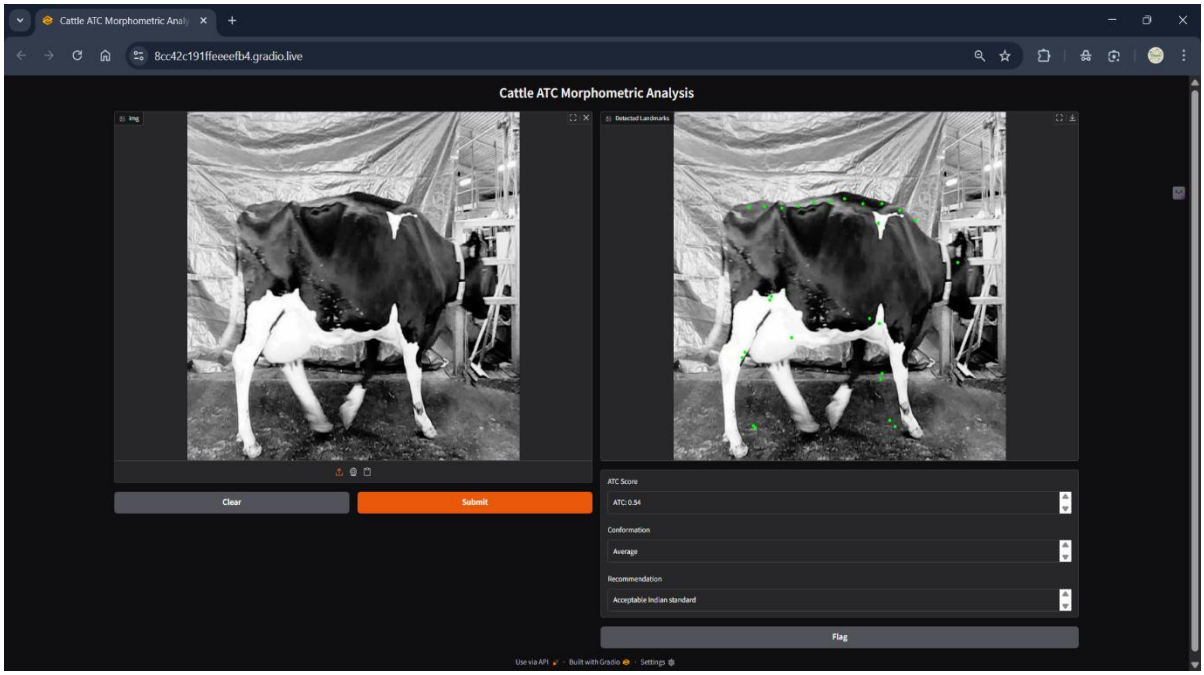

Fig no. 4.7 Snapshot of Output
(Source : Our Code Screenshot)

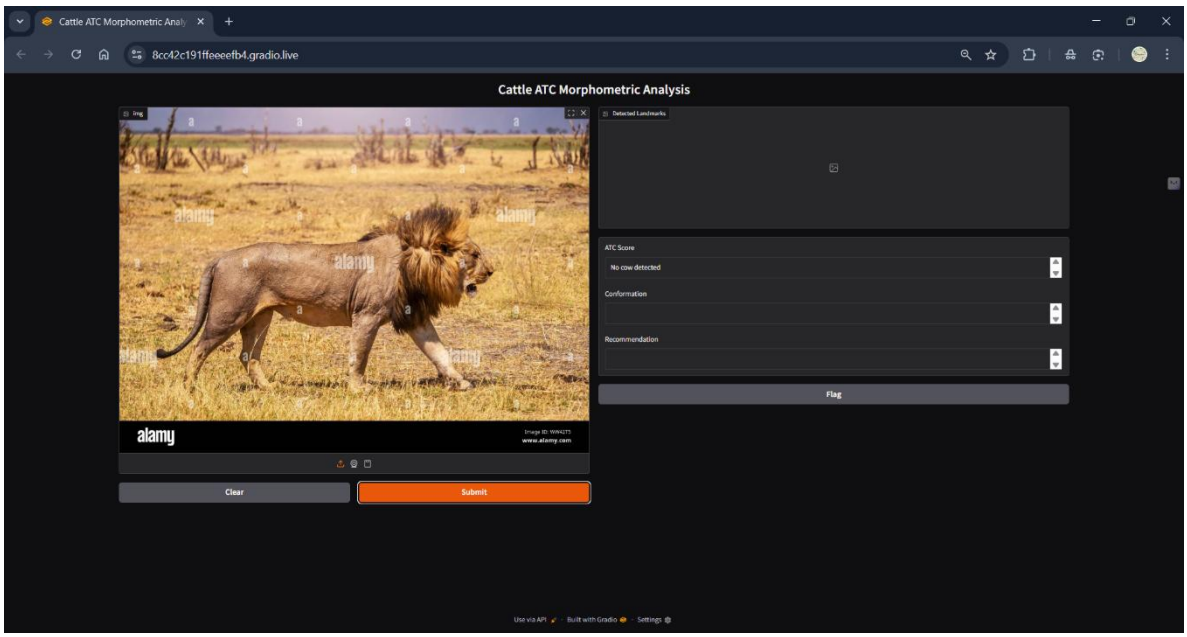

Fig no. 4.8 Snapshot of vision model 2
(Source : Our Code Screenshot)

4.6 DISCUSSION

The experimental result indicates that the combination of anatomical landmark detection and morphometric ratio analysis is an efficient method for automated cattle conformation scoring. The high accuracy of landmark detection indicates that YOLOv8-Pose is an efficient landmark detection method for cattle in a farm environment. The stable morphometric ratios indicate that geometric ratios derived from landmarks are stable and independent of changes in image size.

The variation of ATC scores for cattle images indicates that the system is able to measure significant structural variations, not just average scores. This indicates that the system is valid for cattle conformation measurement using normalized morphometric ratios. In comparison with manual measurement methods, the proposed system does not contain human error and is able to conduct fast structural analysis of cattle images.

The low inference time indicates that the system is able to conduct real-time analysis and can be used for livestock analysis in a real-world environment.

CHAPTER 05
CONCLUSIONS

5.1 SUMMARY OF THE WORK

This work developed an automated image processing system for cattle structural conformation assessment by integrating YOLOv8-Pose landmark localization and morphometric ratio assessment for the evaluation of standardized Animal Type Classification (ATC) scores. Landmarks such as withers, shoulder, hip, and body points were localized and used to estimate body length, chest volume, and rump length, which were ratioed using withers height to obtain scores that were assessed to obtain a composite score for each animal. The system was tested on cattle images obtained in real farm settings and showed accurate landmark localization, reliable measurements, significant variation in ATC scores among animals, and fast processing times, making it an objective automated alternative to manual cattle conformation assessment.

5.2 KEY FINDINGS AND OBSERVATIONS

Project Observations and Key Findings Up to Now To date the observations and key findings of our project are :

- The YOLOv8-Pose model was able to correctly identify cattle anatomical landmarks in the test images.
- The morphometric ratios calculated from the landmarks (OBL/WH, HG/WH, HL/WH) had small variability, thus ensuring geometric consistency in the measurement method.
- The normalized ATC trait scores (Body Length, Chest Capacity, Rump Length) had significant variability, thus ensuring the ability to differentiate structural variations among cattle.
- The ATC index values had variability among the cattle, thus ensuring the ability to rank cattle conformation effectively.
- The average inference time of 10-20 ms per image ensures near real-time processing performance.

5.4 LIMITATIONS OF THE CURRENT WORK

Although the proposed system shows strong performance, there are still some drawbacks.

The morphometric measurements' accuracy is based on the correct identification of the anatomical points. Occlusion, large posture variations, and partial occlusion of the animal

could influence the identification of keypoints and, consequently, the morphometric measurements. The proposed system is based on two-dimensional image geometry and does not consider depth information or perspective distortion.

The test dataset includes a small number of cattle images and does not cover all cattle breeds, ages, and morphologies. Moreover, the ATC scoring system is based on normalized ratio ranges predefined by the system, and not on expert scores or breed-specific standards.

The proposed system processes single side views of the animals and does not consider the combination of multiple views or time-related information that could enhance the accuracy of the morphometric measurements.

5.5 SCOPE FOR FUTURE WORK

The proposed system can be further extended in the following ways in future research :

Using multi-view or video-based images of cattle can improve the localization of landmarks and provide three-dimensional morphometric analysis. The application of depth cameras or stereo vision can provide more accurate structural analysis regardless of the distance of the camera. The addition of different breeds, ages, and farm conditions of cattle to the dataset can improve the generalization of the model and accuracy of scoring.

Future research can also include calibration of ATC scores with expert manual scoring to develop breed-specific standardized conformation scores. Other structural parameters such as leg placement, body depth, and angles can be incorporated to develop full livestock conformation scoring systems.

The application of the system as a mobile or farm-level application can help in its practical implementation in livestock management and breeding programs. The system can be incorporated with dairy farm management systems to provide automated structural analysis and decision support in dairy farming.

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